



Wireless Fixed Camera Network for Greenhouse-Based Plant Phenotyping

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Abstract

An indoor wireless fixed camera network was developed for an efficient, cost-effective method of extracting informative plant phenotypes in a controlled greenhouse environment. Deployed at the Donald Danforth Plant Science Center (DDPSC), this fixed camera platform implements rapid and automated plant phenotyping. The platform uses low-cost Raspberry Pi computers and digital cameras to monitor aboveground morphological and developmental plant phenotypes. The Raspberry Pi is a readily programmable, credit card-sized computer board with remote accessibility. A standard camera module connects to the Raspberry Pi computer board and generates eight-megapixel resolution images. With a fixed array, or “bramble,” of Raspberry Pi computer boards and camera modules placed strategically in a greenhouse, we can capture automated, high-resolution images for 3D reconstructions of individual plants on timescales ranging from minutes to hours, capturing temporal changes in plant phenotypes.

Key words Greenhouse, Phenotyping, Raspberry Pi, 3D reconstruction, Image analysis, Imaging

1 Introduction

A wireless fixed camera array was developed and deployed at the Donald Danforth Plant Science Center (DDPSC) to monitor aboveground morphological and developmental plant phenotypes. One hundred eighty Raspberry Pis (Pis) and connected camera modules (Fig. 1) are positioned on an overhead scaffold in a grid formation. The camera array, or “bramble,” is powered by blocks of electrical power strips with USB adaptors and cables, and network access is provided via WiFi dongles attached to each Pi. The Pis are affixed to the scaffold 3 meters above the ground with flexible tripods (Fig. 2). Each camera is angled to provide overlapping fields of view to ensure high-quality 3D reconstructions.

3D reconstructions and image analyses are carried out in VisualSFM [1] and CloudCompare [2], respectively (Fig. 3). Using the connected component analysis feature in

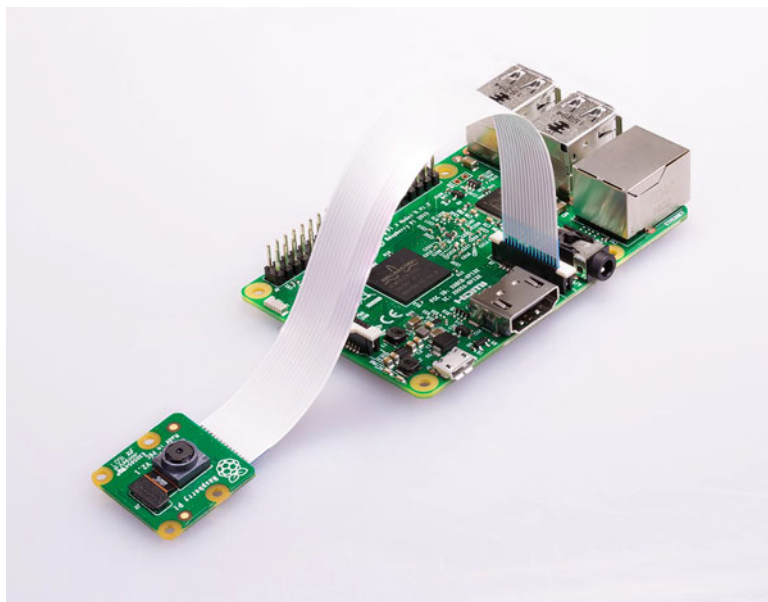


Fig. 1 Raspberry Pi 3 with an attached 8 MP camera module (www.raspberrypi.com)



Fig. 2 Raspberry Pi camera setup (case, WiPi dongle, power cord) from two views attached to scaffold with a flexible tripod

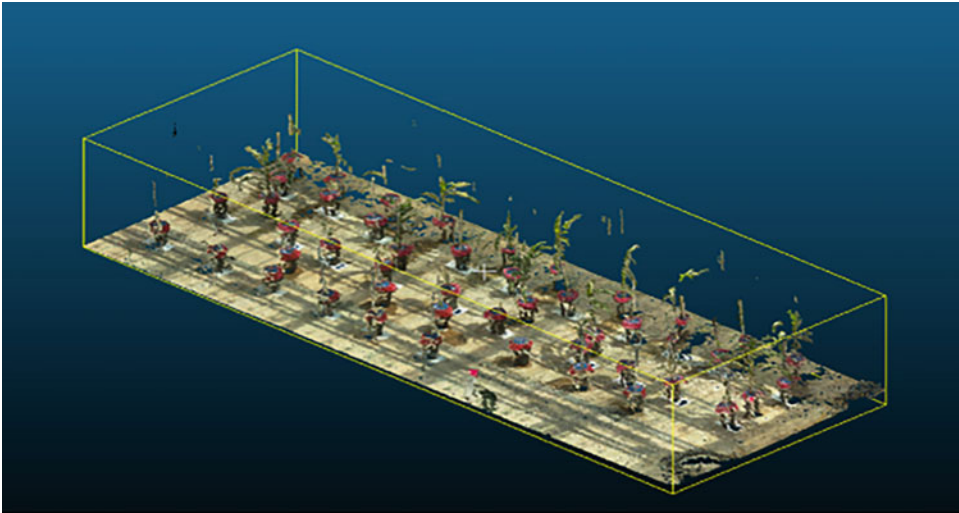


Fig. 3 VisualSFM 3D reconstruction and CloudCompare visualization of the greenhouse point cloud

CloudCompare, we can bound and separate each plant in the greenhouse and accurately calculate measures of biomass accumulation, plant height, leaf area, leaf angle, and growth rate. Comparing ground-truth and image-derived measurements of plant heights, we attain 1–2 cm accuracy in distance measurements with R-squared values of greater than 0.9 (Fig. 4a–c).

2 Materials

- Raspberry Pi 3 computers (*see Note 1*).
- Raspberry Pi 8MP camera modules (*see Note 1*).
- Raspberry Pi camera cases.
- Flexible tripods.
- WiPi WiFi dongles.
- MicroSD cards (minimum 8 GB storage).
- USB power cords.
- Electrical power strips.
- Wireless access point(s)

3 Methods

3.1 Hardware Setup

Raspberry Pis and connected camera modules are enclosed in a water-resistant plastic case and are affixed to an overhead scaffold in an evenly spaced grid formation. WiPi WiFi dongles are used to connect the Pis to the WiFi local area network. Raspberry Pi

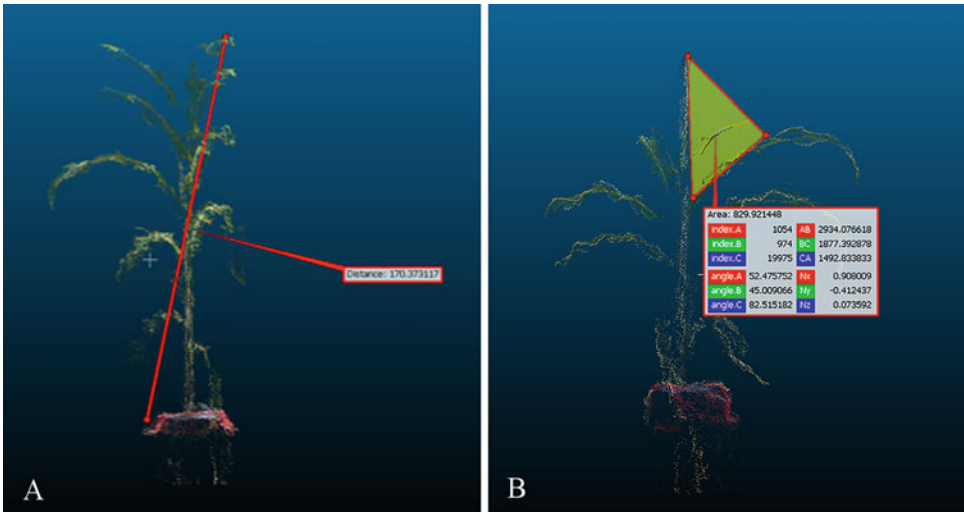


Fig. 4 CloudCompare feature analysis of a single plant with derived plant height (a), leaf angle (b), and leaf lengths (c)

3 computers have internal WiFi; however, we found that using an external WiPi dongle improves signal strength and stability. The camera modules are angled using flexible tripods to ensure 20–30% overlap in the fields of view between adjacent cameras.

3.2 Software and Management

The Raspberry Pi camera array is online at all times and was created using Raspbian, a Debian-based operating system for Raspberry Pi software (www.raspbian.org). Ganglia [3] is used to obtain real-time reporting data from the bramble, including a visualization of the Raspberry Pis that are currently online and transmitting data (Fig. 5). We also recommend spot checking the images downloaded at a given time point or accessing each Pi remotely to verify connectivity and transmission. Independent of network access, image capture is initiated on each Pi camera system via a cron job that is set up on the Raspberry Pi device. Images are stored on a local SD card and then imported to local storage, or a compute infrastructure from each Raspberry Pi using a cron job set up on a rsync server. The Raspberry Pi bramble is managed from the server using a deployment configuration engine called Ansible [4]. Each Pi’s configuration requires a unique hostname and IP address. Additional requirements include WiFi access either via a WiPi dongle or use of Raspberry Pi 3 camera, a set time zone, and an OpenSSH server. A microSD card with a minimum of 8 GB storage is recommended for each Pi system to accommodate system specifications and local image storage.

Wireless power management is a standard feature on the Raspberry Pi that disables the WiFi dongle after periods of inactivity. This setting can be turned off using the code “sudo nano /etc/



Fig. 5 Ganglia interface showing CPU load of each Pi in the Bramble. CPU load for four camera systems are individually highlighted in bottom panels

network/interfaces” and entering “wireless-power off” in the wlan0 section. OpenSSH server is also installed on each Pi using the command “sudo apt-get install openssh-server.” OpenSSH allows for remote access and transfer capabilities. We also recommend setting the time zone for each Pi using the command “sudo dpkg-reconfigure tzdata.” A local image directory is also created using the command “mkdir /home/pi/Images” which creates a directory named “Images” on the Pi.

3.3 Raspberry Pi Bramble Setup

We built an array of 180 Raspberry Pis, which provides sufficient coverage and 20–30% image overlaps for a 1400 sq ft/130 m² greenhouse (25' × 56'/7.6 m × 17.1 m). The Pis are affixed in a grid arrangement to a scaffold (15' × 45'/4.6 m × 13.7 m) 3 m above the ground.

3.4 Calculation of Distance Units

A GPS reference point from the corner of the greenhouse was identified using Google Maps/Earth Mercator projection. QGIS [5] was used to assign GPS points to the Raspberry Pis in the Bramble given the reference GPS location. In VisualSFM, GPS points were added to the 3D reconstruction images, which translates point cloud units into centimeters for visualization and analysis in CloudCompare (Fig. 6).

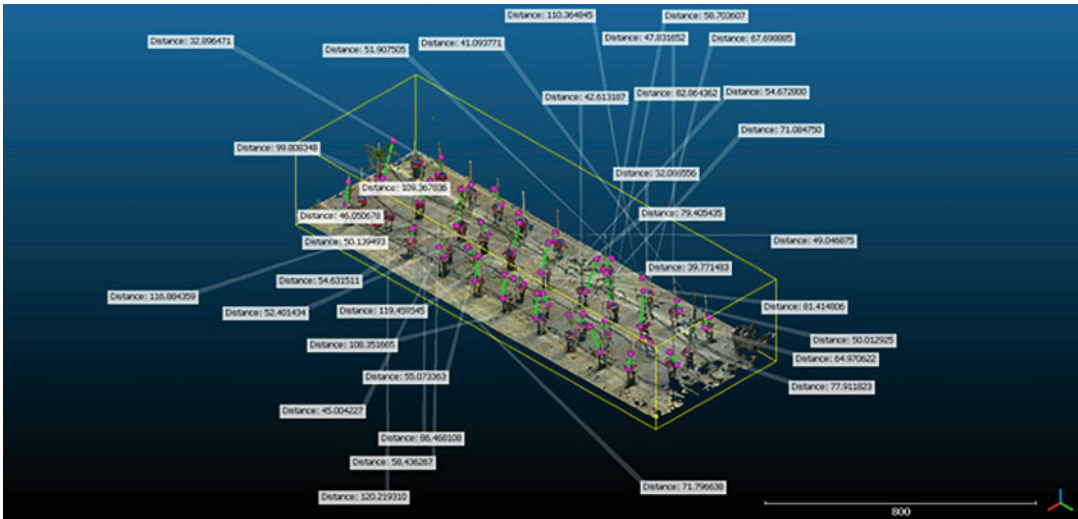


Fig. 6 CloudCompare distance measurements in centimeters

3.5 3D Reconstruction and Analysis in VisualSFM and CloudCompare

VisualSFM is used to reconstruct the interior of the greenhouse, including all plants and pots in the space. Using connected component analysis in CloudCompare, the 3D reconstruction is separated into individual point clouds. Thresholding is used to determine the minimum number of points per component, which subsequently selects the smallest set of points that can be considered a “connected component” (Fig. 7a, b). Complete objects with multiple point clouds (e.g., several leaves on one plant) require manual merging.

4 Notes

1. It is recommended that the newest, highest-resolution Raspberry Pi computers and camera modules available be used when building a camera network.
2. The most significant challenge with the wireless fixed camera system is maintaining a sustained connection to each Raspberry Pi for the amount of time it takes to transfer a file. Wireless interference is the typical cause of this issue. There are four wireless access points (WAPs), each connected to 45 Pis, stationed in and around the greenhouse that cause interference; the 180 Raspberry Pis themselves also cause interference.
3. We implemented several methods to reduce the impact of wireless interference. Four WAPs and four SSIDs were used exclusively for the bramble, preventing Raspberry Pi network dropout. WiPi dongles also helped with establishing connectivity and maintaining connection stability. We also decreased the transmit power from each Raspberry Pi until it

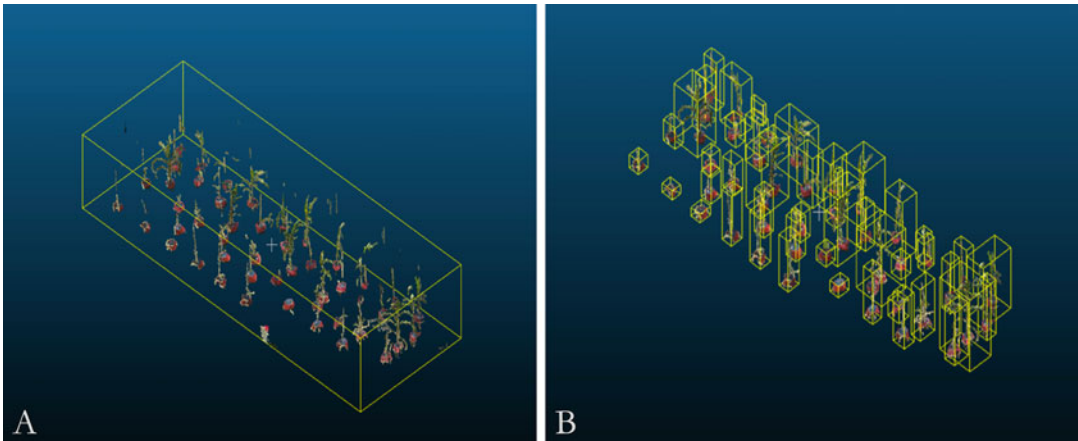


Fig. 7 CloudCompare connected component analysis pipeline with thresholding (a) and connected component extraction (b)

was time to transfer the files. The transmit power was increased for the duration of the transfer time.

4. Hardwiring: If feasible, each Raspberry Pi can be hardwired to a switch that is directly connected to the network.
5. Partial hardwiring: We recommend partially hardwiring the Raspberry 5. Pis, six at a time to a wireless bridge, bringing the number of devices connected to each SSID/WAP to eight Pis per SSID/WAP instead of 45 per SSID/WAP.
6. WiFi dongles with directional antennas: Dongles with directional antennas may also aid in reducing wireless interference.
7. The quality of the Pi-generated input images in VisualSFM determines the quality of the 3D reconstructions. Poor image quality in the greenhouse is primarily due to environmental variables, including variations in sunlight passing through the greenhouse roof and resulting in under- and oversaturated images. Cloudy days or shadows from clouds or overhead lighting fixtures can also dramatically reduce image quality and the resolution of the subsequent 3D reconstruction.
8. Several image processing methods can be applied to the Pi-generated images before the 3D reconstruction step. These include thresholding, machine learning-based classification methods, and image enhancements [6–8]. Addition of color markers or other distinct objects in the greenhouse can also aid in the reconstruction process. We have found that VisualSFM produces a better reconstruction output when there are unique features captured in the greenhouse during image acquisition. Additionally, we find that increasing the rate of image acquisition allows for the removal of sub-par 3D reconstructions without compromising the phenotyping experiment.

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References

1. Wu C (2011) VisualSFM: a visual structure from motion system
2. CloudCompare (version 2.1) (2019) Retrieved from <https://www.cloudcompare.org/>
3. Massie ML, Chun BN, Culler DE (2004) The ganglia distributed monitoring system: design, implementation, and experience. *Parallel Comput* 30(7):817–840
4. Hochstein L, Moser R (2017) Ansible: up and running: automating configuration management and deployment the easy way. O'Reilly Media, Inc
5. QGIS Development Team. (2016) QGIS geographic information system. Open Source Geospatial Foundation Project. Retrieved from <https://www.osgeo.org/>
6. Yu K, Kirchgessner N, Grieder C et al (2017) An image analysis pipeline for automated classification of imaging light conditions and for quantification of wheat canopy cover time series in field phenotyping. *Plant Methods* 13(1):15
7. Liebisch F, Kirchgessner N, Schneider D et al (2015) Remote, aerial phenotyping of maize traits with a mobile multi-sensor approach. *Plant Methods* 11(1):9
8. Hartmann A, Czauderna T, Hoffmann R et al (2011) HTPheno: an image analysis pipeline for high-throughput plant phenotyping. *BMC Bioinform* 12(1):148